

11.5-11.8

11.5 Relativistic perturbations to hydrogen

kinetic energy but relativistic

Thus the first-order shift in the energy due to the relativistic correction to the electron's kinetic energy is given by

$$E_K^{(1)} = -\frac{1}{2}m_e c^2 Z^4 \alpha^4 \left[-\frac{3}{4n^4} + \frac{1}{n^3(l + \frac{1}{2})} \right] \quad (11.74)$$

$$\hat{H}_{S-O} = \frac{Ze^2}{2m_e^2 c^2 |\hat{\mathbf{r}}|^3} \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} \quad (11.78)$$

Why does the Dirac equation have four components instead of two? Any quantum mechanical relativistic description of particles must include the antiparticles as well as the particles—in this case positrons as well as electrons. *Each* of these particles is a spin- $\frac{1}{2}$ particle, and thus we end up with a four-component equation. Why must the positrons be included in our treatment? One way to see this is to go back to the energy-time uncertainty relation (4.63) and note that for time intervals

$$\Delta t \sim \frac{\hbar}{m_e c^2} \quad (11.100)$$

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fine, hyperfine, relativistic effects, and quantum electrodynamics.

11.7 Zeeman in hydrogen

$$\hat{H}_B = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = -\left(\frac{-e}{2m_e c} \hat{\mathbf{L}} + \frac{-e}{m_e c} \hat{\mathbf{S}} \right) \cdot \mathbf{B} \quad (11.104)$$

we orient in the z direction, evaluate energy

Hence

$$E_B^{(1)} = \frac{e\hbar B}{2m_e c} m_j \left(1 \pm \frac{1}{2l+1} \right) \quad (11.109)$$

Notice that we can express this energy shift compactly in the form

$$E_B^{(1)} = \frac{g(j, l)e\hbar B}{2m_e c} m_j \quad (11.110)$$

which is reminiscent of the energy of a particle of spin j in an external magnetic field with a g factor

$$g(j = l \pm \frac{1}{2}, l) = \left(1 \pm \frac{1}{2l+1} \right) \quad (11.111)$$

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$$E_n^{(1)} = \langle \varphi_n^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle \quad (11.114)$$

and

$$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle \varphi_k^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}. \quad (11.115)$$